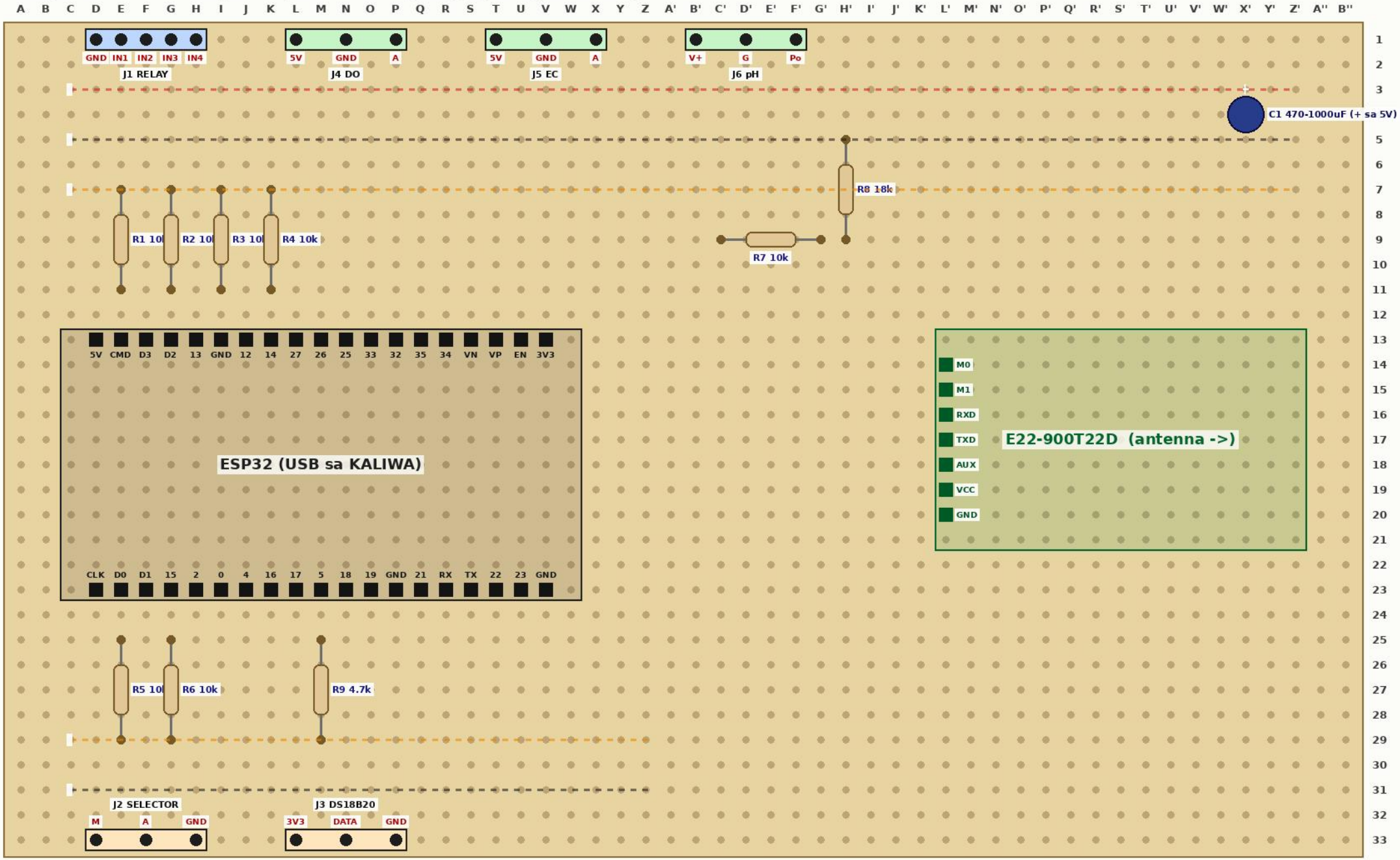
TOP VIEW - component side (dito ilalagay ang parts)

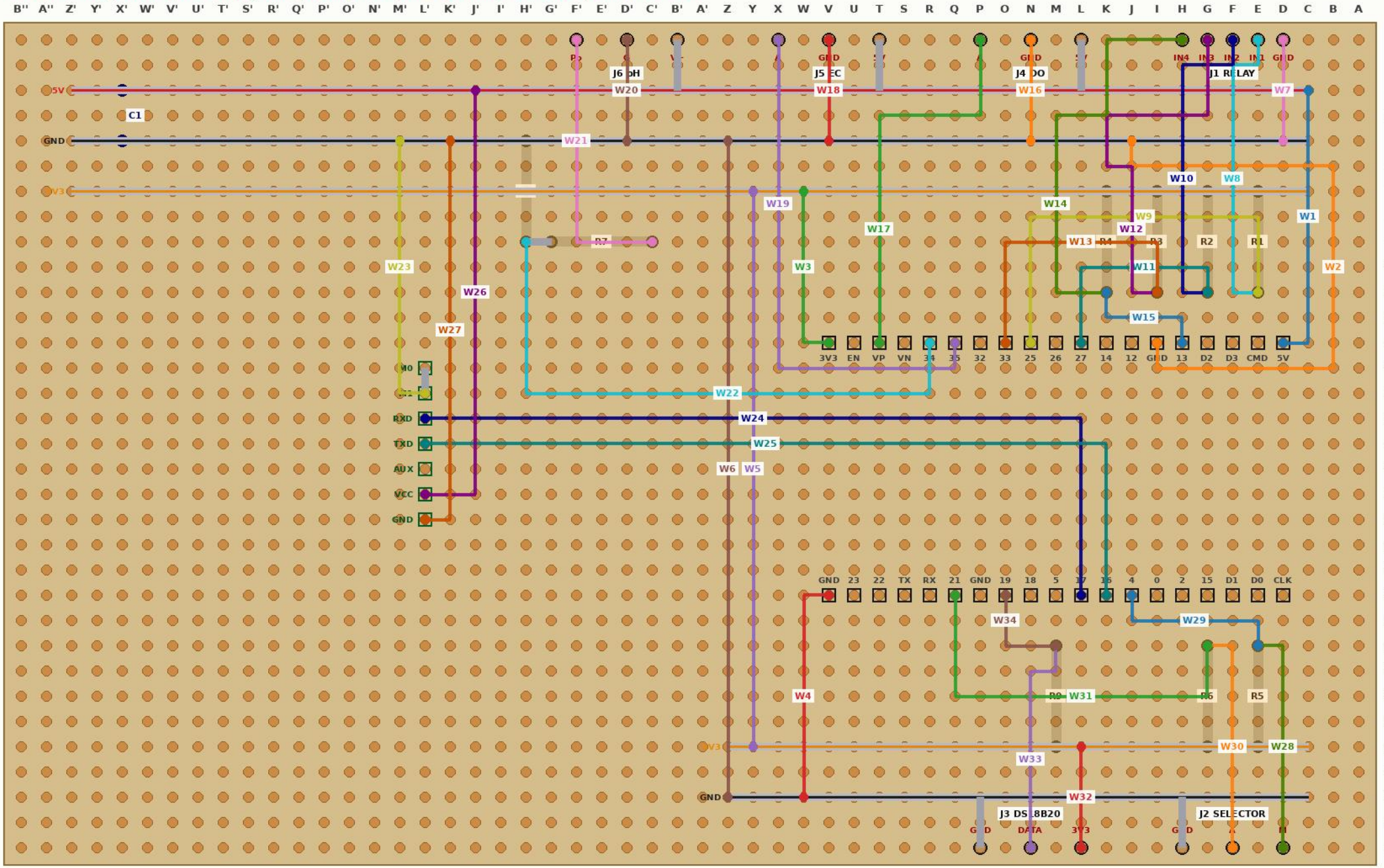
Tinitingnan mula sa ITAAS. Walang wire dito; nasa ilalim lahat ng wiring (tingnan ang BOTTOM VIEW).



Putol-putol na linya = rail sa ILALIM (gabay lang). Rails: row 3 = 5V, row 5 = GND, row 7 = 3V3, row 29 = 3V3 (B), row 31 = GND (B)

BOTTOM VIEW - solder side (dito ang rails at wires)

NAKA-MIRROR: ganito ang itsura kapag binaligtad ang board (kaliwa-pakanan). Column A ay nasa KANAN na.



bare tinned wire (rails at maiikling tulay) — insulated wire W# (tingnan ang wire list) — Bilog/kuwadrado na walang laman = lead/pin galing sa itaas

Perfboard Build Guide - Parts, Rails, Bridges

Board: dot perfboard 54 x 33 (2.54 mm). Format ng posisyon: row,column (hal. 13,D).
Columns: A-Z, saka A'-Z' (pangalawang A-Z), saka A'' at B''. Row 1 = itaas.

1. Mga parts (component side / ITAAS)

Part	Halaga / uri	Posisyon
ESP32 socket	2x 19-pin female header	row 13 at row 23, columns D-V (USB sa kaliwa)
E22 socket	1x 7-pin female header	column L', rows 14-20 (M0 sa itaas)
J1 Relay	1x 5-pin female header	row 1, D-H: GND IN1 IN2 IN3 IN4
J4 DO	3P screw terminal 5.08	row 1: L=5V N=GND P=A
J5 EC	3P screw terminal 5.08	row 1: T=5V V=GND X=A
J6 pH	3P screw terminal 5.08	row 1: B'=V+ D'=G F'=Po
J2 Selector	3P screw terminal 5.08	row 33: D=M F=A H=GND
J3 DS18B20	3P screw terminal 5.08	row 33: L=3V3 N=DATA P=GND
R1	10k ohm, 1/4 W	7,E -> 11,E
R2	10k ohm, 1/4 W	7,G -> 11,G
R3	10k ohm, 1/4 W	7,I -> 11,I
R4	10k ohm, 1/4 W	7,K -> 11,K
R7	10k ohm, 1/4 W	9,C' -> 9,G'
R8	18k ohm, 1/4 W	5,H' -> 9,H'
R5	10k ohm, 1/4 W	25,E -> 29,E
R6	10k ohm, 1/4 W	25,G -> 29,G
R9	4.7k ohm, 1/4 W	25,M -> 29,M
C1	470-1000 uF, >=16 V electrolytic	+ sa 3,X' (5V) - sa 5,X' (GND)

2. Rails (bare tinned wire sa ILALIM)

Rail	Row	Columns
5V	3	C hanggang Z'
GND	5	C hanggang Z'
3V3	7	C hanggang Z'
3V3 (B)	29	C hanggang Z
GND (B)	31	C hanggang Z

3. Maiikling bare bridges (sa ILALIM)

```
[ ] 1,L  ->  3,L                    DO 5V -> 5V rail
[ ] 1,T  ->  3,T                    EC 5V -> 5V rail
[ ] 1,B'  ->  3,B'                  pH V+ -> 5V rail
[ ] 9,G'  ->  9,H'                  pH divider node (R7 + R8)
[ ] 14,L'  ->  15,L'                E22 M0 + M1
[ ] 33,H  ->  31,H                  Selector GND -> GND rail B
[ ] 33,P  ->  31,P                  DS18B20 GND -> GND rail B
```

Mga paalala

- Rails at bridges lang ang bare wire. Lahat ng iba ay INSULATED wire (22-24 AWG).
- Ang insulated wire ay puwedeng tumawid sa ibabaw ng rails nang walang short.
- Kapag 2-3 bagay ang nagkikita (hal. J1 IN1 + GPIO25 + R1), i-solder sa lead ng resistor.
- Walang anumang 12 V o 220 V sa perfboard na ito.
- Ang VCC ng relay module ay galing pa rin sa buck OUT+, HINDI sa perfboard.

Perfboard Build Guide - Insulated Wires (W1-W34)

Sundin ang ruta sa BOTTOM VIEW. Lagyan ng tsek ang bawat wire kapag tapos na.

	Wire	Mula	Papunta	Para saan
[ ]	W1	13,D	3,C	ESP32 5V -> 5V rail
[ ]	W2	13,I	5,J	ESP32 GND -> GND rail
[ ]	W3	13,V	7,W	ESP32 3V3 -> 3V3 rail
[ ]	W4	23,V	31,W	ESP32 GND -> GND rail B
[ ]	W5	7,Y	29,Y	3V3 rail -> 3V3 rail B
[ ]	W6	5,Z	31,Z	GND rail -> GND rail B
[ ]	W7	1,D	5,D	J1 GND -> GND rail
[ ]	W8	1,E	11,E	J1 IN1 -> R1
[ ]	W9	13,N	11,E	GPIO25 -> R1 (IN1)
[ ]	W10	1,F	11,G	J1 IN2 -> R2
[ ]	W11	13,L	11,G	GPIO27 -> R2 (IN2)
[ ]	W12	1,G	11,I	J1 IN3 -> R3
[ ]	W13	13,O	11,I	GPIO33 -> R3 (IN3)
[ ]	W14	1,H	11,K	J1 IN4 -> R4
[ ]	W15	13,H	11,K	GPIO13 -> R4 (IN4)
[ ]	W16	1,N	5,N	DO GND -> GND rail
[ ]	W17	1,P	13,T	DO A -> VP (GPIO36)
[ ]	W18	1,V	5,V	EC GND -> GND rail
[ ]	W19	1,X	13,Q	EC A -> GPIO35
[ ]	W20	1,D'	5,D'	pH G -> GND rail
[ ]	W21	1,F'	9,C'	pH Po -> R7
[ ]	W22	9,H'	13,R	pH divider node -> GPIO34
[ ]	W23	15,L'	5,M'	E22 M0/M1 -> GND rail
[ ]	W24	23,L	16,L'	GPIO17 -> E22 RXD
[ ]	W25	23,K	17,L'	GPIO16 -> E22 TXD
[ ]	W26	19,L'	3,J'	E22 VCC -> 5V rail
[ ]	W27	20,L'	5,K'	E22 GND -> GND rail
[ ]	W28	33,D	25,E	Selector M -> R5
[ ]	W29	23,J	25,E	GPIO4 -> R5 (M)
[ ]	W30	33,F	25,G	Selector A -> R6
[ ]	W31	23,Q	25,G	GPIO21 -> R6 (A)
[ ]	W32	33,L	29,L	DS18B20 3V3 -> 3V3 rail B
[ ]	W33	33,N	25,M	DS18B20 DATA -> R9
[ ]	W34	23,O	25,M	GPIO19 -> R9 (DATA)

Perfboard Build Guide - Pagkakasunod at Testing

Bago magsimula (i-verify sa mismong board)

[ ] DOT perfboard (hiwa-hiwalay na pads), hindi strip.
[ ] Ipatong ang ESP32: 10 rows ang agwat ng dalawang header? Kung 9, gawing row 22 ang bottom header.
[ ] Basahin ang pin labels ng ESP32 at E22; dapat tugma sa TOP VIEW.
[ ] Screw terminals ay 5.08 mm (bawat ikalawang butas).

Pag-solder

1. Rails: 5 bare wire sa ilalim (rows 3, 5, 7, 29, 31).
2. TEST: beep 5V rail <-> GND rail <-> 3V3 rail: WALANG beep sa pagitan nila.
3. Female headers ng ESP32 (row 13 at 23) at E22 (L'). Huwag munang isaksak ang modules.
4. Resistors R1-R9 (tingnan ang halaga!) at C1 (+ sa 5V rail, row 3).
5. Connectors J1-J6.
6. Bare bridges (7 pirasong maiikli).
7. Insulated wires W1-W34, ayon sa wire list at BOTTOM VIEW.
8. Linisin ang flux at tingnan kung may tumalsik na solder sa pagitan ng pads.

Testing (WALANG nakasaksak na ESP32 at E22)

[ ] Beep: 5V vs GND, 5V vs 3V3, 3V3 vs GND -> WALANG beep
[ ] Beep: J1 IN1 <-> header pin ng GPIO25 (13,N) -> may beep (ganito rin IN2/27, IN3/33, IN4/13)
[ ] Beep: J4 A <-> VP (13,T); J5 A <-> 35 (13,Q) -> may beep
[ ] Ohm: J6 Po <-> GPIO34 (13,R) = ~10k; GPIO34 <-> GND = ~18k
[ ] Beep: J2 M <-> GPIO4 (23,J); J2 A <-> GPIO21 (23,Q); J3 DATA <-> GPIO19 (23,O)
[ ] Beep: E22 RXD <-> GPIO17 (23,L); E22 TXD <-> GPIO16 (23,K)
[ ] Ohm: bawat IN1-IN4 <-> 3V3 rail = ~10k; J3 DATA <-> 3V3 = ~4.7k
[ ] Isaksak ang ESP32 (USB laptop lang). Sukatin: 5V rail ~5 V, 3V3 rail ~3.3 V
[ ] I-upload ang relay_test_k1k4.ino, saka selector_test.ino -> pareho ang resulta sa breadboard

Kung may mali

- May beep sa pagitan ng rails -> may solder bridge; hanapin bago isaksak ang ESP32.
- Walang beep sa signal -> hindi naka-solder ang dulo ng wire, o mali ang butas.
- Mainit ang C1 o baligtad ang + -> hugutin agad; + dapat sa 5V rail.